

# Assignment 3 - Discrete Mathematics (Monsoon 2025)

*“A great discovery solves a great problem but there is a grain of discovery in any problem.”*  
- by George Pólya

**Submission deadline:** 11:59 PM on 23rd November, 2025.

**Maximum marks:** 40

**Instructions.** Consider the following while preparing the answers for the given questions.

1. Kindly go over the academic dishonesty policy of IIIT Delhi given on [this link](#). Any deviation from this policy will lead to the strictest possible punishment.
2. Each question is of 10 marks.
3. It is prohibited to use AI-related tools like chatGPT to solve the questions, unless explicitly mentioned. Any violation in this regard will be handled very strictly.
4. You are encouraged to write solutions in L<sup>A</sup>T<sub>E</sub>X. Anyone who submits homework in L<sup>A</sup>T<sub>E</sub>X (i.e., in the pdf format and not the source code) will get an additional 20% of their score in that homework.
5. While writing the solutions, you can refer to the topics, theorems, examples, etc., taught in the class. It is not allowed to refer to anything related to propositional and predictive logic not covered in the class.
6. Late submissions are not allowed.
7. If you discuss anything related to the solution of a question in this assignment with anyone, you must mention their names.

## 1 Problem Set

**Question 1.1** (10 marks). *Let  $A$  be the set of all finite sequences in  $\mathbb{N}$ . Is  $A$  countably infinite or uncountable? Prove your answer.*

*A finite sequence in  $\mathbb{N}$  is a sequence having finitely many natural numbers. For example,  $(1, 2, 3)$ ,  $(9, 100, 0, 5)$ ,  $(1)$ .*

### Solution and Rubric:

We show that  $A$  is countably infinite.

We treat sequences of length  $k \geq 1$ . (If one includes the empty sequence, the conclusion remains the same.)

For each  $k \geq 1$ , define

$$A_k = \mathbb{N}^k = \{(x_1, \dots, x_k) : x_i \in \mathbb{N}\}.$$

Then

$$A = \bigcup_{k=1}^{\infty} A_k.$$

Our proof has three steps:

1. Show  $\mathbb{N} \times \mathbb{N}$  is countably infinite.
2. Use induction to show  $\mathbb{N}^k$  is countably infinite for all  $k \geq 1$ .
3. Show the countable union  $\bigcup_{k=1}^{\infty} \mathbb{N}^k$  is countably infinite.

**Claim 1:**  $\mathbb{N} \times \mathbb{N}$  is countably infinite.

**Proof:** Enumerate pairs by increasing sum  $i + j$ , breaking ties by the first coordinate:

$$(0, 0), (0, 1), (1, 0), (0, 2), (1, 1), (2, 0), (0, 3), (1, 2), (2, 1), (3, 0), \dots \quad (1)$$

Let  $\pi : \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$  map  $n$  to the  $(n+1)$ -th pair in (1). Every pair appears exactly once, so  $\pi$  is bijective. Thus  $\mathbb{N} \times \mathbb{N}$  is countably infinite.  $\square$

**Claim 2:** Each  $A_k = \mathbb{N}^k$  is countably infinite.

**Proof:** We use induction on  $k$ .

**Base Case ( $k = 1$ ):**  $\mathbb{N}^1 = \mathbb{N}$  is countably infinite.

**Induction Hypothesis:** Assume  $\mathbb{N}^n$  is countably infinite. Thus there exists a bijection

$$f^{-1} : \mathbb{N}^n \rightarrow \mathbb{N}.$$

(We keep the notation  $f^{-1}$  as in the original draft.)

Also from Claim 1 we have a bijection

$$g^{-1} : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}.$$

**Inductive Step ( $n \rightarrow n+1$ ):** We define a map

$$h : \mathbb{N}^{n+1} = \mathbb{N}^n \times \mathbb{N} \rightarrow \mathbb{N}$$

by

$$h((x_1, \dots, x_n), y) = g^{-1}(f^{-1}(x_1, \dots, x_n), y).$$

We now prove  $h$  is bijective by proving injectivity and surjectivity directly (no composition notation).

**Injectivity:** Suppose

$$h((x_1, \dots, x_n), y) = h((x'_1, \dots, x'_n), y').$$

By definition of  $h$  this means

$$g^{-1}(f^{-1}(x_1, \dots, x_n), y) = g^{-1}(f^{-1}(x'_1, \dots, x'_n), y').$$

Since  $g^{-1}$  is injective (it is a bijection), we must have

$$(f^{-1}(x_1, \dots, x_n), y) = (f^{-1}(x'_1, \dots, x'_n), y').$$

Thus  $y = y'$  and

$$f^{-1}(x_1, \dots, x_n) = f^{-1}(x'_1, \dots, x'_n).$$

Because  $f^{-1}$  is injective, it follows that

$$(x_1, \dots, x_n) = (x'_1, \dots, x'_n).$$

Therefore

$$((x_1, \dots, x_n), y) = ((x'_1, \dots, x'_n), y'),$$

so  $h$  is injective.

**Surjectivity:** Let  $m \in \mathbb{N}$  be arbitrary. Since  $g^{-1}$  is surjective, there exist  $p, q \in \mathbb{N}$  such that

$$g^{-1}(p, q) = m.$$

Because  $f^{-1}$  is surjective, there exists  $(x_1, \dots, x_n) \in \mathbb{N}^n$  with

$$f^{-1}(x_1, \dots, x_n) = p.$$

Then by definition of  $h$ ,

$$h((x_1, \dots, x_n), q) = g^{-1}(f^{-1}(x_1, \dots, x_n), q) = g^{-1}(p, q) = m.$$

Thus every  $m \in \mathbb{N}$  is attained by  $h$ , so  $h$  is surjective.

Having shown  $h$  is both injective and surjective,  $h$  is bijective. Therefore  $\mathbb{N}^{n+1}$  is countably infinite, completing the inductive step.  $\square$

**Claim 3:**  $A = \bigcup_{k=1}^{\infty} A_k$  is countably infinite.

Since each  $A_k$  is countably infinite, fix bijections

$$\varphi_k : \mathbb{N} \rightarrow A_k.$$

Arrange all elements in an infinite table:

$$\begin{array}{cccc}
A_1 : & \varphi_1(0) & \varphi_1(1) & \varphi_1(2) & \cdots \\
A_2 : & \varphi_2(0) & \varphi_2(1) & \varphi_2(2) & \cdots \\
A_3 : & \varphi_3(0) & \varphi_3(1) & \varphi_3(2) & \cdots \\
\vdots & \vdots & \vdots & \vdots & \vdots
\end{array}$$

Enumerate elements by diagonal traversal of index pairs  $(k, j)$ :

$$(1, 0), (1, 1), (2, 0), (1, 2), (2, 1), (3, 0), (1, 3), (2, 2), (3, 1), (4, 0), \dots$$

Let  $F : \mathbb{N} \rightarrow A$  be the function sending  $n$  to the  $(n + 1)$ -th element in this diagonal listing.

**Surjectivity:** Any element of  $A$  belongs to some  $A_k$ , say  $a = \varphi_k(j)$ . The pair  $(k, j)$  occurs in the diagonal enumeration, so  $a = F(n)$  for some  $n$ .

**Injectivity:** Distinct pairs  $(k, j)$  correspond to distinct elements  $\varphi_k(j)$  because:

- if  $k \neq k'$ , the sequences have different lengths;
- if  $k = k'$  but  $j \neq j'$ , injectivity of  $\varphi_k$  gives different elements.

Thus  $F$  is injective.

Since  $F$  is bijective,  $A$  is countably infinite. □

**Conclusion:** The set  $A$  of all finite sequences of natural numbers is **countably infinite**.

**Rubric (10 Marks)**

- **1 mark:** Correctly states that  $A$  is countably infinite.
- **2 marks:** Defines  $A_k = \mathbb{N}^k$  and partitions  $A$  based on sequence length.
- **2 marks:** Gives a correct proof that  $\mathbb{N} \times \mathbb{N}$  is countably infinite.
- **2 marks:** Provides a complete induction proof that  $\mathbb{N}^k$  is countably infinite.
- **3 marks:** Gives a rigorous proof that  $A = \bigcup_{k=1}^{\infty} \mathbb{N}^k$  is countably infinite using diagonal enumeration.

**Question 1.2.** 1. (5 marks). Let  $n \in \mathbb{N} - \{0\}$  and  $A$  be the set of all 0,1 sequences containing  $n$  bits. Let  $R$  be a relation on  $A$  defined as follows: for every  $(a_1, a_2, \dots, a_n), (b_1, b_2, \dots, b_n) \in A, ((a_1, \dots, a_n), (b_1, \dots, b_n)) \in R$  if and only if

$$a_1 + \dots + a_n = b_1 + \dots + b_n.$$

Prove that  $R$  is an equivalence relation. Count the number of equivalence classes in  $A/R$ . Describe an equivalence class in  $A/R$  that has the maximum number of elements among all equivalence classes in  $A/R$ , and count the number of elements present in this equivalence class.

*Solution and Rubric:*

### 1. Proof of Equivalence

- **Reflexive:** For any sequence  $a$ ,  $\sum a_i = \sum a_i$ . Thus  $(a, a) \in R$ .
- **Symmetric:** If  $\sum a_i = \sum b_i$ , then  $\sum b_i = \sum a_i$ . Thus  $(b, a) \in R$ .
- **Transitive:** If  $\sum a_i = \sum b_i$  and  $\sum b_i = \sum c_i$ , then  $\sum a_i = \sum c_i$ . Thus  $(a, c) \in R$ .

### 2. Number of Equivalence Classes

The sum of bits for a sequence of length  $n$  can range from 0 (all zeros) to  $n$  (all ones). The set of possible sums is  $\{0, 1, \dots, n\}$ .

- **Answer:** There are  $n + 1$  equivalence classes.

### 3. Maximum Equivalence Class

The size of a class is maximized when the number of 1s is closest to half the length  $(n/2)$ .

- **Description:** Sequences with  $\lfloor n/2 \rfloor$  ones.

- **Count:**  $\binom{n}{\lfloor n/2 \rfloor}$ .

**Rubric:**

- 2 marks for proof of equivalence
- 1.5 marks for count of classes
- 1.5 max class details

2. (5 marks). Let  $A = \{1, \dots, n\}$  for  $n \in \mathbb{N} - \{0\}$ . Let  $B$  be all permutations on  $A$ . Define the equivalence relation  $R$  on  $B$  as follows: for every  $f, g \in B$ ,  $(f, g) \in R$  if and only if there exist  $\pi, \sigma \in B$  such that  $f = \sigma \circ g \circ \pi$ .

- (a) Let  $[f]_R$  be an arbitrary equivalence class in  $B/R$ . Count the number of elements in  $[f]_R$ .  
 (b) Count the number of equivalence classes in  $B/R$ . Justify your answers.

**Solution and Rubric:**

Context: The relation is defined on the set  $B$  of all permutations  $(S_n)$ , where

$$f \sim g \iff f = \sigma \circ g \circ \pi$$

for some  $\sigma, \pi \in B$ . This allows pre-composition and post-composition by arbitrary permutations.

- **(a) Size of Class  $[f]_R$**  Since we can choose any  $\pi$  and  $\sigma$ , this relation allows mapping any bijective function  $g$  to any other bijective function  $f$  (it forms the Universal Relation on the set). Therefore, the equivalence class  $[f]_R$  contains every element of  $B$ .

Count:  $n!$  elements.

- **(b) Number of Equivalence Classes** Since the single equivalence class contains the entire set of permutations, there are no other disjoint classes.

Count: 1 class.

**Rubric:**

- 2.5 marks for size of class
- 2.5 marks for number of classes

**Question 1.3.** 1. (5 marks). Let  $\{A_i : i \in \mathbb{N}\}$  be the set of countably infinite sets, i.e., for every  $i \in \mathbb{N}$ ,  $A_i$  is countably infinite. Prove that for every  $n \in \mathbb{N}, n \geq 1$ ,  $A_1 \cup \dots \cup A_n$  and  $A_1 \cap \dots \cap A_n$  are countable.

**Solution and Rubric:**

Let  $\{A_i : i \in \mathbb{N}\}$  be a collection of countably infinite sets. That is, for each  $i \in \mathbb{N}$ , there exists a bijection

$$f_i : \mathbb{N} \rightarrow A_i.$$

We will prove that for every  $n \in \mathbb{N}, n \geq 1$ :

- the union  $A_1 \cup \dots \cup A_n$  is countable, and
- the intersection  $A_1 \cap \dots \cap A_n$  is countable.

**(A) Countability of finite unions (3 marks)**

Let

$$B = A_1 \cup A_2 \cup \dots \cup A_n.$$

Since each  $A_i$  is countably infinite, there exists a surjection  $f_i : \mathbb{N} \rightarrow A_i$  for  $i = 1, \dots, n$ .

Define a function

$$g : \mathbb{N} \times \{1, 2, \dots, n\} \rightarrow B, \quad g(k, i) = f_i(k).$$

Claim:  $\mathbb{N} \times \{1, 2, \dots, n\}$  is countable.

Reason:  $\mathbb{N}$  is countable, and  $\{1, \dots, n\}$  is finite. The Cartesian product of a countable set with a finite set is countable. Hence  $\mathbb{N} \times \{1, \dots, n\}$  is countable.

The image of  $g$  is

$$\text{Im}(g) = B.$$

Thus  $B$  is the image of a countable set under a function, so  $B$  is at most countable:

$$|A_1 \cup \dots \cup A_n| \leq |\mathbb{N} \times \{1, \dots, n\}| = \aleph_0.$$

Therefore  $A_1 \cup \dots \cup A_n$  is countable.

**(B) Countability of finite intersections (2 marks)**

Let

$$C = A_1 \cap A_2 \cap \dots \cap A_n.$$

Then clearly  $C \subseteq A_1$ .

Since  $A_1$  is countably infinite, it is countable. A basic result is that any subset of a countable set is countable (or finite). Therefore  $C$  is countable.

Hence

$$A_1 \cap \dots \cap A_n \text{ is countable.}$$

**Conclusion:** For every  $n \geq 1$ , both the finite union  $A_1 \cup \dots \cup A_n$  and the finite intersection  $A_1 \cap \dots \cap A_n$  of countably infinite sets are countable.

**Rubric:**

• **Finite union is countable (3 marks):**

- (1 mark) Correctly states or uses that each  $A_i$  is countable and corresponds to a map  $f_i : \mathbb{N} \rightarrow A_i$ .
- (1 mark) Constructs or references a countable set such as  $\mathbb{N} \times \{1, \dots, n\}$  and defines a suitable function  $g$ .
- (1 mark) Correctly concludes that  $A_1 \cup \dots \cup A_n$  is countable as the image of a countable set.

• **Finite intersection is countable (2 marks):**

- (1 mark) Observes  $A_1 \cap \dots \cap A_n \subseteq A_1$  (or any  $A_i$ ).
- (1 mark) Uses the fact that a subset of a countable set is countable.

2. (5 marks). Prove the following for every  $n \in \mathbb{N}, n \geq 1$ : Let  $A$  be a set of  $n$  elements and  $B$  be the set of all subsets of  $A$  containing even number of elements, i.e.,

$$B = \{X \subseteq A : |X| \text{ is even}\}.$$

Then,  $|B| \leq 2^{n-1}$ .

**Solution and Rubric:**

Let  $A$  be a set with  $|A| = n \geq 1$  and define

$$B = \{X \subseteq A : |X| \text{ is even}\}.$$

We prove by induction on  $n$  that  $|B| \leq 2^{n-1}$ , and in fact  $|B| = 2^{n-1}$ .

**Base Case.** Let  $n = 1$ . Then  $A = \{x\}$ . The subsets are  $\emptyset$  and  $\{x\}$ . The only subset with even cardinality is  $\emptyset$ , so

$$B = \{\emptyset\}, \quad |B| = 1 = 2^0 = 2^{n-1}.$$

**Induction Hypothesis.** Assume that for a set  $A'$  with  $|A'| = k$ , the collection

$$B' = \{X \subseteq A' : |X| \text{ is even}\}$$

satisfies

$$|B'| = 2^{k-1}.$$

**Induction Step.** Let  $A$  be a set with  $|A| = k + 1$ . Choose  $x \notin A'$  so that

$$A = A' \cup \{x\}, \quad |A'| = k.$$

Every subset of  $A$  falls into one of the following categories:

- **Case 1:** Subsets not containing  $x$ . These are exactly subsets of  $A'$ , giving  $|B'|$  even subsets.
- **Case 2:** Subsets containing  $x$ . Such a subset is  $Y \cup \{x\}$ , where  $Y \subseteq A'$ . This has even size iff  $|Y|$  is odd.

Let

$$C' = \{Y \subseteq A' : |Y| \text{ is odd}\}.$$

Since  $2^k = |B'| + |C'|$ , we have

$$|C'| = 2^k - |B'|.$$

Thus the total number of even subsets of  $A$  is

$$|B| = |B'| + |C'| = |B'| + (2^k - |B'|) = 2^k.$$

But for  $n = k + 1$ , we have  $2^{n-1} = 2^k$ . Hence

$$|B| = 2^{n-1}.$$

By induction,  $|B| = 2^{n-1}$  for all  $n \geq 1$ .

- 1 mark for correctly identifying and proving the base case.
- 1 mark for clearly writing the induction hypothesis.
- 2 marks for the correctness and clarity of the induction step, including the two-case split and proper use of the hypothesis.
- 1 mark for the correct final conclusion stating the result holds for all  $n \geq 1$  by induction.

**Question 1.4.** 1. (4 marks). Let  $a_1 := 1$  and for every  $n \geq 2$ , let  $a_n := n - a_{n-1}$ . For every  $n \in \mathbb{N}, n \geq 1$ , prove that  $a_n \geq 1$ .

*Solution and Rubric:*

**Solution:**

- **Claim:** For every integer  $n \geq 1$  we have  $a_n \geq 1$ , where  $a_1 := 1$  and for  $n \geq 2$ ,

$$a_n := n - a_{n-1}.$$

**Base case:**

$n = 1$ : By definition  $a_1 = 1$  so  $a_1 \geq 1$ .

$n = 2$ :

$$a_2 = 2 - a_1 = 2 - 1 = 1.$$

Thus  $1 \leq a_2 \leq 2 - 1$ . So the statement holds for  $n = 1, 2$ .

**Induction Hypothesis:** Assume for some integer  $k \geq 2$  that

$$1 \leq a_k \leq k - 1.$$

**Inductive step:**

Assume  $a_k \geq 1$  for all  $k \leq n$ ,  $n \geq 2$ .

For

$$a_{n+1} = (n + 1) - a_n,$$

we need  $a_{n+1} \geq 1$ , i.e.,

$$(n + 1) - a_n \geq 1 \iff a_n \leq n.$$

But

$$a_n = n - a_{n-1},$$

and  $a_{n-1} \geq 1$  by IH, so

$$a_n \leq n - 1,$$

which is  $\leq n$ . So  $a_n \leq n$  is true.

Thus,

$$a_{n+1} = (n+1) - a_n \geq (n+1) - n = 1.$$

Hence  $a_{n+1} \geq 1$ .

**Rubric:**

- Claim clearly stated — 1 mark
- Base Case — 0.5 mark
- Induction Hypothesis — 1 mark (If hypothesis is wrong: 0 marks and no evaluation of inductive step)
- Inductive Step / Proof — 1.5 mark

2. (6 marks). Write a summary of this course, preferably in the form of a story. Highlight your learnings. Your answer should be contained in at most two pages.

**Rubric:**

No rubric for this question as it is a subjective to TA's interpretation.